# Lab 07: VLAN Segmentation

2960-24TT Switch · PC0 · PC1 · PC2 – Three VLANs

**Lab #07** **Topic:** VLANs & Switching **Tool:** Cisco Packet Tracer **Cert:** CompTIA Network+

## 1. Lab Objectives

- Create and name three VLANs on a single Cisco 2960-24TT switch.
- Assign PC0, PC1 and PC2 to different VLANs via access ports.
- Verify that devices in the same VLAN can communicate at Layer 2.
- Confirm that devices in different VLANs are isolated (no routing).
- Relate to Network+ objective 2.3: Purpose and use of VLANs.

## 2. Network Topology

A single 2960-24TT switch (Switch0) is the only network device. PC0 connects on Fa0/1, PC1 on Fa0/2, PC2 on Fa0/3. Each PC is in a different VLAN, so without a router or Layer 3 switch they are fully isolated.

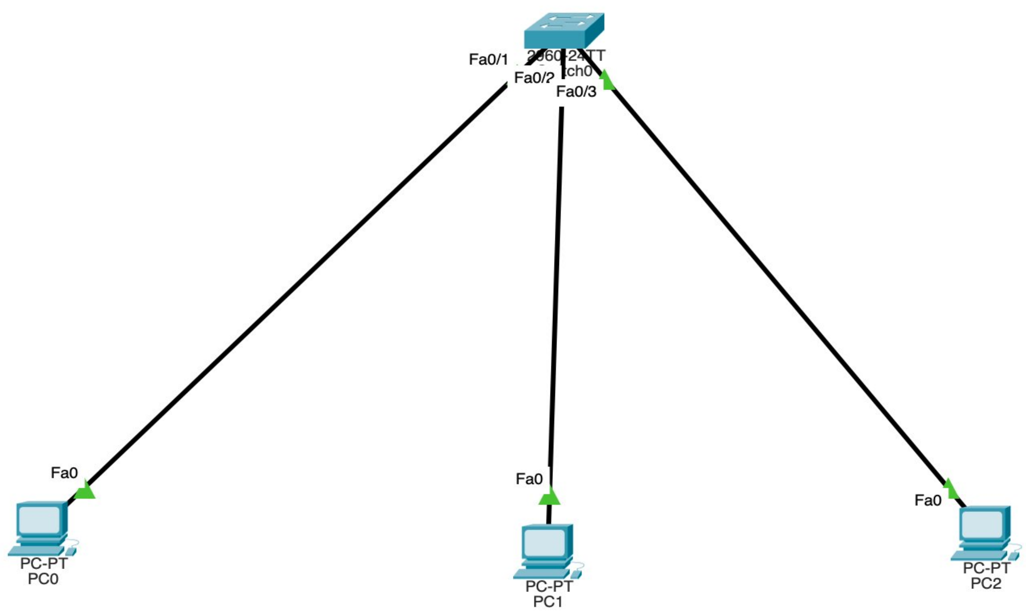


*Figure 1 – Single switch with PC0, PC1, PC2 on Fa0/1, Fa0/2, Fa0/3*

| Device | Switch Port | VLAN ID | VLAN Name | IP Address | Subnet Mask |
|---|---|---|---|---|---|
| PC0 | Fa0/1 | VLAN 10 | SALES | 192.168.10.10 | 255.255.255.0 |
| PC1 | Fa0/2 | VLAN 20 | TECH | 192.168.20.10 | 255.255.255.0 |

| | | | | | |
|---|---|---|---|---|---|
| PC2 | Fa0/3 | VLAN 30 | MGMT | 192.168.30.10 | 255.255.255.0 |

## 3. Step-by-Step Configuration

### 3.1 Create VLANs

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name SALES
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name TECH
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name MGMT
Switch(config-vlan)# exit
```

### 3.2 Assign Ports to VLANs

```
Switch(config)# interface FastEthernet0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
Switch(config)# interface FastEthernet0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)# interface FastEthernet0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# exit
Switch# write memory
```

## 4. Verification

```
Switch# show vlan brief          ! Lists all VLANs and port assignments
PC0> ping 192.168.10.x           ! Within VLAN 10 — SUCCESS
PC0> ping 192.168.20.10          ! Cross VLAN — FAIL (no routing)
PC0> ping 192.168.30.10          ! Cross VLAN — FAIL (no routing)
```

## 5. Key Takeaways

- VLANs divide a physical switch into multiple logical broadcast domains.
- Access ports carry traffic for only one VLAN — no tagging on the frame.
- Without a Layer 3 device, hosts in different VLANs cannot communicate at all.
- VLANs improve security, reduce broadcast traffic and simplify network management.
- VLAN IDs are locally significant — they must match between switches on trunk links.